

CONDITIONAL VISUO-TEXTUAL PROMPT LEARNING FOR MEDICAL IMAGE ANALYSIS

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ABSTRACT

The paper presents CTVTL-CLIP, a conditional visuo-textual prompt tuning method designed to adapt large multimodal models for highly specialized applications, where the required semantic representations are not initially encoded in the models’ latent space. CTVTL-CLIP jointly optimizes CLIP’s text and visual encoders using sets of soft prompts that condition the representations on domain-specific knowledge and features, with the support of a lightweight pretrained network to condition textual learnable tokens. This approach enhances the model’s ability to accurately understand medical images, aligning textual descriptions with visual representations even in cases with limited annotated data. CTVTL-CLIP was tested on an endoscopy image dataset containing multiple gastric lesions, significantly outperforming traditional classifiers such as CNNs and vision transformers, while using fewer learnable parameters. It was also evaluated on two additional medical image analysis tasks, including skin lesion classification and stenosis classification in angiographies, showing improved performance compared to state-of-the-art methods that require training more parameters. The combination of CTVTL-CLIP’s efficiency and superior performance underscores its practical potential for real-world medical applications, where data scarcity and model efficiency are critical challenges.

Index Terms— vision-language models, soft prompt tuning, medical imaging analysis

1. INTRODUCTION

The convergence of computer vision and natural language processing has significantly reshaped artificial intelligence research in recent years, largely driven by the introduction of transformer-based architectures. These models, such as those popularized by Vaswani et al. [1] and Jia et al. [2], have enabled substantial advancements across multiple tasks, including image classification, object detection, and semantic segmentation. Transformer models like GPT [3] leverage self-attention mechanisms to capture intricate relationships within

both visual and textual data, marking a departure from traditional methods that treated these modalities separately.

A key breakthrough in vision-language research has been CLIP (Contrastive Language-Image Pre-training) [4], a model that learns rich, transferable representations by aligning images and text within a shared feature space. By doing so, CLIP can generalize across a variety of tasks without task-specific fine-tuning, providing a versatile tool for applications requiring cross-domain scalability. The impact of models like CLIP and ALIGN [2] has been profound, setting new standards for handling visual and textual data by utilizing large-scale datasets and contrastive learning methods. However, CLIP’s versatility can be constrained in domain-specific scenarios, such as medical imaging, where understanding nuanced features is essential for accurate interpretation.

Efforts to address these limitations have led to the development of specialized foundation models tailored for medical applications [5–8]. Nevertheless, fine-tuning transformer-based models for medical imaging often demands extensive labeled data and computational resources, which may not be readily available. As a more accessible alternative, prompt tuning methods—such as visual prompt tuning (VPT) [9] or textual prompt tuning [10, 11] and related techniques in natural language processing [12–16]—have emerged, enabling models to learn either text or visual prompts with minimal adjustments to parameters. To further address this gap, we propose Conditional Text and Visual Tokens Learnable in CLIP (CTVTL-CLIP), a novel approach that enhances CLIP’s performance in medical image classification by simultaneously optimizing textual and visual embeddings with soft prompts, that excels in challenging medical tasks while also reducing computational demands.

2. METHOD

The proposed methodology builds upon the foundational architecture of the CLIP model, introducing enhancements to boost its performance and adaptability to domain-specific datasets. We augment the CLIP model with learnable tokens in both the textual and visual encoders, inspired by recent advancements in vision-language research and prompt learning

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methodologies.

Let $\mathbb{V}$ be a text vocabulary, and let $\mathbb{V}^N$ denote a sequence of words from $\mathbb{V}$; for simplicity of notation, we assume N to be “flexible” (i.e., varying as appropriate), as transformer-based text encoders are designed to be length-agnostic, through the usage of padding tokens. Let $\varepsilon_t : \mathbb{V}^N \rightarrow \mathbb{R}^{N \times d}$ be a text embedding function, projecting each vocabulary word to the embedding size d . A text encoder $\mathcal{T} : \mathbb{R}^{N \times d} \rightarrow \mathbb{R}^d$ (e.g., from CLIP) is pretrained to project a text sequence to a numeric representation of size d . In our work, we assume to know the names of the K target classes in text format, i.e., $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_K$, with $\mathbf{c}_i \in \mathbb{V}^N$. We then introduce a set P of textual *class-specific context* prompts (CSC), $P = \{\mathbf{p}_{1,1}, \dots, \mathbf{p}_{1,M}, \mathbf{p}_{2,1}, \dots, \mathbf{p}_{2,M}, \dots, \mathbf{p}_{K,1}, \dots, \mathbf{p}_{K,M}\}$, with M learnable prompts for each class. When extracting the text encoding for class k through $\mathcal{T}$, instead of simply computing $\mathcal{T}(\varepsilon(\mathbf{c}_k))$, we concatenate the class-specific prompts to the embedded class name, obtaining $\mathcal{T}([\mathbf{p}_{k,1}, \dots, \mathbf{p}_{k,M}, \varepsilon_t(\mathbf{c}_k)])$. The motivation for this kind of conditioning lies in the observation that the text encoder may not know the correct semantics of domain-specific terms, especially if they have different meanings in common usage (e.g., the word “erosion”, which denotes a kind of lesion in gastroenterology). In this way, we allow the text encoder to adapt its representations in a parameter-efficient fashion, learning to extract more discriminative feature of the target concept. A simpler variant of this approach does not employ *class-specific* tokens, but only employs a single set of M tokens that are concatenated to each class name. However, we identified a key limitation in class-specific context (CSC) learnable text prompts: as the number of classes increases, parameter complexity scales with $O(2^{n.classes})$, leading to exponential growth in training time. To address this challenge, we focused on use only non-CSC text prompts with the addition of *visual contextual features* CT . These features are extracted from a shallow frozen *support network* which effectively distinguishes efficiently low-level image context.

A similar enhancement is applied for visual feature extraction. Let $\mathcal{I}$ be the image space and $\varepsilon_v : \mathcal{I} \rightarrow \mathbb{R}^{L \times d}$ a visual embedding function, which converts an image into M patches and projects them to a d -dimensional embedding space. Let $\mathcal{V} : \mathbb{R}^{(L+1) \times d} \rightarrow \mathbb{R}^d$ be a visual encoder, which receives a sequence of M patch embeddings, appends a learnable CLS token and uses the corresponding output features as a representative vector of the image, lying on the same embedding space as text encoder $\mathcal{T}$. Moreover, let $\mathcal{V}_i : \mathbb{R}^{(L+1) \times d} \rightarrow \mathbb{R}^{(L+1) \times d}$ be the i -th layer of $\mathcal{V}$, such that $\mathcal{V}(\mathbf{i}) = \text{extr}(\mathcal{V}_H(\mathcal{V}_{H-1}(\dots(\mathcal{V}_1(\mathbf{i}))))$, with $\mathbf{i} \in \mathcal{I}$, H being the total number of layers and extr the function that selects the CLS token output. As above, we introduce learnable tokens for the visual encoder as well. In the most general formulation, we employ a set $V = \{\mathbf{v}_{1,1}, \dots, \mathbf{v}_{1,Q}, \mathbf{v}_{2,1}, \dots, \mathbf{v}_{2,Q}, \dots, \mathbf{v}_{H,1}, \dots, \mathbf{v}_{H,Q}\}$, i.e., with H groups of Q tokens for each layer of visual encoder $\mathcal{V}$.

The encoder’s output is then computed as $\mathcal{V}(V, \varepsilon_v(\mathbf{i}))$, where internally each layer computes $\mathcal{V}_i(\text{repQ}(\mathbf{v}_{i,1}, \dots, \mathbf{v}_{i,Q}, \mathcal{V}_{i-1}))$, with $\mathcal{V}_{i-1}$ used for short to represent the previous layer’s output, and repQ indicating that the first Q tokens of the previous layer are replaced with $\mathbf{v}_{i,1}, \dots, \mathbf{v}_{i,Q}$.

At training time, we optimize the values of the learnable prompts from P and V to minimize the following loss function:

$$\mathcal{L} = -\mathbb{E}_{\mathbf{i}, y} [\mathcal{T}([\mathbf{p}_{y,1}, \dots, \mathbf{p}_{y,M}, \varepsilon_t(\mathbf{c}_y)]) \cdot \mathcal{V}(V, \varepsilon_v(\mathbf{i}))] \quad (1)$$

with y being the correct class index for input $\mathbf{i}$. In practice, we intend to maximize the dot product between the visual representation of an image and the textual representation corresponding to the correct class.

The reason for adopting the *deep prompting* approach in the visual encoder is to perform a stronger conditioning on the entire network, which needs to be adapted to become able to recognize visual patterns from a domain which may differ significantly from the data used at pretraining time. Such strong conditioning is not necessary for the text encoder, since it “only” needs to provide the weights to be used for linear classification. It should also be noted that the visual encoder does not perform class-specific prompting, since of course the corresponding class label is unknown to the visual encoder. A simpler variant of the above approach employs a smaller set of learnable visual tokens $V = \{\mathbf{v}_1, \dots, \mathbf{v}_Q\}$, reused through all encoder layers.

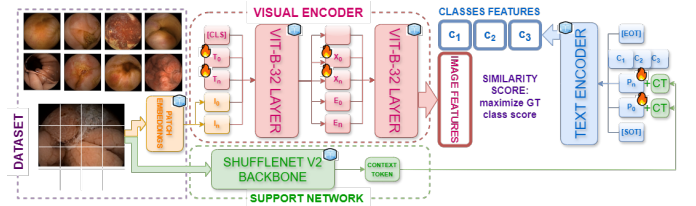


Fig. 1: CTVTL-CLIP architecture. The support network produces a context token that is added to the learnable tokens. These, along with the labels, are fed into the text encoder to generate embeddings for each class. The correct class should minimize the similarity score with the visual embedding outputted by the visual encoder (ViT-B-32), which takes learnable patches and visual patch embeddings as input.

3. EXPERIMENTAL RESULTS

CTVTCL-CLIP is first tested on a private dataset consisting of gastroscopy images collected by means of video capsule endoscopy. The dataset encompasses a diverse range of gastrointestinal pathologies and was collected from 2018 to 2021 at the University Hospital of Catania in Italy. It includes data from 713 patients, aged between 43 and 79 years, comprising 317 females and 396 males, all of Italian nationality and

residents of the southern regions of Italy.

The data collection involved patients ingesting a capsule containing two cameras, which continuously recorded at VGA resolution (640x480) at 15 FPS. From the collected videos, two expert gastroendoscopy doctors identified and annotated multiple images.

The dataset comprises a total of 1,842 images from gastroscopic examinations, annotated into seven distinct pathological classes plus negative samples: negatives (700 images), erosion (229 images), bleeding (56 images), petechial (47 images), polyp (67 images), ulcer (76 images), angiodysplasia (490 images), hyperemia (132 images).

The training methodology for CTVTL-CLIP was designed to optimize the performance trade-off between sensitivity and specificity in gastrointestinal pathology classification tasks while ensuring computational efficiency and scalability. Optimization of the visual and text encoders was carried out for 200 epochs through stochastic gradient descent with a learning rate of 0.002 and a batch size of 128. For computational efficiency, we employed 16-bit floating point training using CUDA Automatic Mixed Precision. To address class imbalance, we employed a weighted variant of the loss function in Eq. 1, with weights determined based on the occurrence percentage of each class in the training set. By default, in our experiments we use a ShuffleNet V2 backbone as support network, a ViT-B-32 network as the visual encoder and a text encoder as described in [4]. We first start our analysis by identifying the best configuration in terms of text, visual and class-specific tokens. Table 1 presents the performance comparison across various configurations of text and visual tokens evaluated for the CTVTL-CLIP model. Each configuration is characterized by the number of text tokens (*Text Token*), whether a class-specific context was used (*CSC*), w/o context token contribution (*CT*) from support network, the number of visual tokens added (*Visual Token*), and whether visual tokens were added at each visual transformer layer (*Deep Visual Token*). Results illustrate that configurations with 64 textual tokens (*non-CSC*, with *CT* from support network) and 5 deep visual tokens (*Visual Token* set to 5, *Deep Visual Token* set to ✓) achieved the highest accuracy balanced of 89.4%, with good improvements in sensitivity (80.3%), specificity (98.6%), and F1 score (80.8%). These findings underscore the importance of support network context and deep visual representations in enhancing model performance for gastrointestinal pathology classification.

We then compare the performance of CTVTL-CLIP against several state-of-the-art architectures, including DenseNet, RegNet, ResNet, SqueezeNet, Swin Transformer, ViT, and related variants, both with and without pre-training, as shown in Table 2. CTVTL-CLIP achieves an accuracy balanced of **89.4%**, which represents the highest accuracy among all models considered. Our model exhibits a slightly lower sensitivity of **80.3%**, while it stands out on distinguishing negative cases (a specificity of **98.6%**). Moreover, CTVTL-

Table 1: Performance comparison with different combinations of text and visual tokens.

Text Token	CSC	Visual Token	Deep V. Token	Acc. Bal.	Sens.	Spec.	F1 Score
0		0		48.2	10.6	85.8	10.2
0		5	✓	84.2	70.8	97.6	77.9
1	✗	0		54.6	11.5	97.7	17.6
1	✓	0		54.6	11.5	97.7	17.6
64	✓	0		76.9	57.6	96.2	68.4
64	✗	5	✗	78.6	60.0	97.2	66.0
64	✗	5	✓	86.8	75.4	98.1	78.2
64	✓	5	✗	76.7	55.5	97.9	68.2
64 +CT	✗	5	✗	80.0	62.3	97.7	69.6
64	✓	5	✓	88.7	79.2	98.3	80.2
64 +CT	✗	5	✓	89.4	80.3	98.6	80.8

CLIP is highly efficient, requiring training only about 80,000 parameters, significantly fewer than its competitors.

Thus, our proposed method yields the best trade-off between specificity, sensitivity, and efficiency (less learnable parameters), which are crucial for accurate diagnosis of gastrointestinal lesions. Specificity ensures the model’s ability to correctly identify true negatives, minimizing false alarms and unnecessary procedures for patients without lesions. Sensitivity ensures the model can detect all true positives, essential for identifying lesions requiring immediate medical attention.

Table 2: Comparison with methods from the state of the art.

Model	# learnable params ($\times 10^6$)	Acc. Bal.	Sens.	Spec.	F1 Score
DenseNet-161	26.49	85.9	83.6	82.9	81.9
↪ Pre-trained		83.8	87.9	83.3	79.7
ResNet-50	23.52	73.5	78.0	66.1	63.0
↪ Pre-trained		82.2	93.5	73.5	77.6
SqueezeNet	0.73	79.3	40.7	81.7	61.1
↪ Pre-trained		85.6	75.2	86.4	76.1
ViT-B-32	87.46	76.5	38.8	80.9	50.6
↪ Pre-trained		80.9	60.7	83.7	66.3
CTVTL-CLIP	0.08	89.4	80.3	98.6	80.8

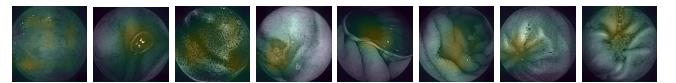


Fig. 2: The attention maps corresponding to each class (none, erosion, bleeding, petechial, polyp, ulcer, angiodysplasia, hyperemia) are displayed.

We integrated the Transformer-MM-Explainability technique [17] into our framework to enhance interpretability. Transformer-MM-Explainability generates attention maps, visually depicting where the model focuses during its decision-making process. Examples of attention maps for each class obtained by CTVTL-CLIP are shown in Fig. 2.

Once the optimal configuration of text and visual prompts

is identified, we assess the generalization capability of our proposed approach by extending its application to two additional medical imaging datasets:

- **FFR**: an extended version of the dataset presented in [18], consisting of 1,001 angiographic x-ray images. Two different classification tasks are performed: binary task for hemodynamically significant stenoses, multi-class categorization for classifying into 3 level of severity.
- **Skin Lesion Pathology**: we employ the ISIC 2019 challenge dataset [19–21], which includes 25,331 skin images across eight diagnosis categories: Melanoma, Melanocytic nevus, Basal cell carcinoma, Actinic keratosis, Benign keratosis, Dermatofibroma, Vascular lesion, and Squamous cell carcinoma.

In Table 3, we compare the results achieved by our approach on these three datasets with those by a ViT-B-32 model (same of CLIP visual encoder), pre-trained on ImageNet and fine-tuned on these tasks. Our approach consistently outperforms the baseline, showing that parameter-efficient adaptation of visual features, combined with a learnable text-based representation of the target concepts, can outperform a fully-tuned ViT-B-32 model on domain-specific tasks.

Table 3: Results on additional medical datasets. Comparison between a ViT-B-32 baseline and our best model.

Dataset	Acc. Bal.	Sens.	Spec.	F1 Score
ViT-B-32 baseline (pretr. on ImageNet, fine-tuned on each dataset)				
FFR Intervention	66.5	66.1	66.9	61.6
FFR Stenosis Severity	69.2	51.6	86.8	63.9
Skin Lesion Pathology	80.6	72.1	89.1	72.7
Proposed approach				
FFR Intervention	72.1	69.3	74.9	72.4
FFR Stenosis Severity	79.5	63.2	95.8	74.7
Skin Lesion Pathology	84.9	74.6	95.2	76.2

In our ablation study, we conduct an investigation into the impact of learnable textual/visual tokens and the support network on the performance of CTVTL-CLIP for gastrointestinal pathology detection. Our experiments focus on isolating and analyzing the individual contributions of each modality by testing configurations where only learnable textual tokens (w/wo context token) or only learnable visual tokens are incorporated into CLIP architecture.

In the “Text Tokens Only” configuration, visual tokens are excluded from the model architecture, underscoring the single contribution of learnable text token provided to the CLIP text encoder. Table 4 shows that performance gradually increases with the number of learnable tokens and when introducing CSC tokens, but still exhibits a significant drop compared to our full model. This suggests that adapting the text encoder with learnable soft prompts is promising for domain-specific classifiers but is less effective without corresponding modifications to visual representations.

Table 4: Performance using only learnable text tokens (*Text Tok.*: n. textual tokens, *CSC*: w/wo class-specific context).

Text Tok.	CSC	Acc. Bal.	Sens.	Spec.	F1 Score
0		48.2	10.6	85.8	10.2
1	✗	54.6	11.5	97.7	17.6
1	✓	54.6	11.5	97.7	17.6
1 +CT	✗	56.8	16.1	97.5	21.1
16	✗	67.9	37.4	98.3	45.8
16	✓	67.9	37.4	98.3	45.8
16 +CT	✗	74.0	51.2	96.8	55.1
64	✗	76.9	57.6	96.2	68.4
64	✓	76.9	57.6	96.2	68.4
64 +CT	✗	77.2	57.7	96.7	68.8
Ours		89.4	80.3	98.6	80.8

Table 5: Results using only learnable visual token (*Vis. Tok.*: n. visual tokens, *Deep Vis. Tok.*: w/wo layer’s visual tokens).

Vis. Tok.	Deep Vis. Tok.	Acc. Bal.	Sens.	Spec.	F1 Score
0		48.2	10.6	85.8	10.2
1	✗	50.9	5.9	95.8	6.8
1	✓	65.4	33.5	97.3	40.1
5	✗	51.6	6.0	97.3	6.5
5	✓	84.2	70.8	97.6	77.9
10	✗	54.5	13.0	96.0	17.8
10	✓	82.8	68.0	97.5	72.8
Ours		89.4	80.3	98.6	80.8

Conversely, in the “Visual Tokens Only” configuration, learnable text tokens are omitted, and the model solely relies on learnable visual tokens. As is shown in Tab. 5, the adoption of visual soft prompts is critical for achieving high accuracy, even without having to include too many learnable tokens. It is even more interesting to notice that the main contribution to classification accuracy is due to the inclusion of deep layer-specific tokens, showing that a strong conditioning is required to adapt visual features to novel domain-specific problems, for which the visual encoder has little knowledge.

Furthermore, we also experimented with other architectures for the *support network*, such as RegNet and MaxViT w/wo pretraining and n-layer MLP, without achieving improved results.

4. CONCLUDING REMARKS

In this work, we have developed and evaluated CTVTL-CLIP on multiple image analysis tasks by leveraging advanced vision-language pre-training techniques. By effectively aligning text and visual modalities within a unified feature space, and incorporating automated prompt engineering techniques, we significantly enhance model adaptability across diverse datasets and clinical scenarios, enhancing efficiency and reducing computational costs. Additionally, we are investigating federated learning applications to enhance data privacy and scalability, crucial for collaborative medical research.

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